

TOPIC

07

Acids, bases and salts

Questions only • Original examination pages retained

24

QUESTIONS

347

TOTAL MARKS

1

SECTIONS



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01

Acids, bases, salts and pH

24 questions • 347 marks

7.1 Acids, bases, salts and pH

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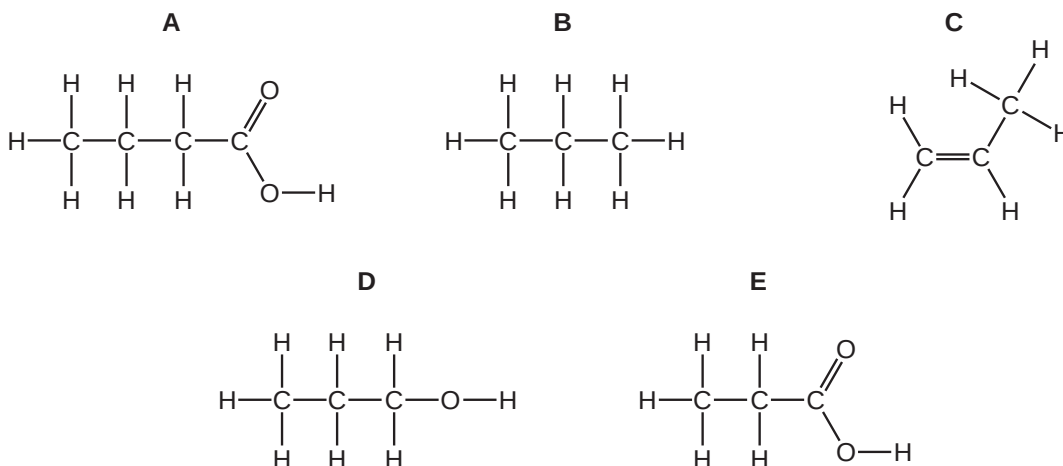
7.1 Acids, bases, salts and pH

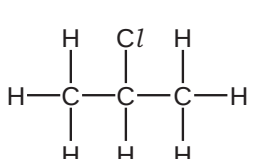
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- 1 (a) The structures of five organic compounds, **A**, **B**, **C**, **D** and **E**, are shown.

Answer the questions that follow.

Each letter may be used once, more than once or not at all.



- (i) Give the letter of the compound that is propan-1-ol.
..... [1]
- (ii) Give the letter of the compound that has the empirical formula CH_2 .
..... [1]
- (iii) Give the letter of **one** compound that reacts with bromine in an addition reaction.
..... [1]
- (iv) Give the letter of **one** compound that reacts with chlorine to form the compound shown.
- 
- [1]
- (v) Give the letters of **two** compounds that can react with each other to form an ester.
..... and [1]
- (vi) Give the letter of the compound that is in the same homologous series as hex-1-ene.
..... [1]
- (vii) Give the letter of **one** compound that is an acid.
..... [1]

(viii) Draw a structural isomer of compound D.

Show all of the atoms and all of the bonds.

[1]

(b) Some acids are described as weak acids.

State the meaning of the term *weak acid*.

weak

acid

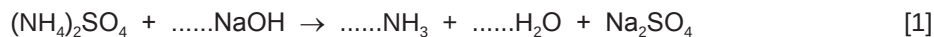
[2]

[Total: 10]

- 3 (a) Aqueous ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$, is warmed with aqueous sodium hydroxide.

The pungent-smelling gas ammonia, NH_3 , is produced.

Balance the equation for this reaction.



- (b) A 2.8 g sample of impure ammonium sulfate is found to contain 0.7 g of impurities.

Calculate the percentage of ammonium sulfate in this sample.

percentage of ammonium sulfate = % [1]

- (c) Describe a test for ammonia gas.

test

result [2]

- (d) Ammonia gas is prepared at the front of a laboratory.

The pungent smell of ammonia spreads throughout the laboratory slowly.

- (i) Name the process that occurs when ammonia gas spreads throughout the laboratory.

..... [1]

- (ii) Explain, using ideas about particles, why ammonia gas spreads throughout the laboratory.

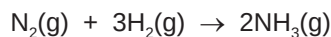
.....
.....
.....
..... [2]

- (iii) Explain why carbon dioxide gas, CO_2 , will spread throughout the laboratory at a slower rate than ammonia gas, NH_3 .

.....
..... [1]

- (e) Ammonia is produced in the Haber process.

The equation for the reaction is shown.



- (i) In the Haber process, a temperature of 450°C and a pressure of 200 atmospheres are used in the presence of finely-divided iron.

A larger equilibrium yield of ammonia would be produced if a lower temperature and a higher pressure are used.

Explain why a lower temperature and a higher pressure are **not** used.

lower temperature

.....

higher pressure

.....

[2]

- (ii) State the role of iron in the Haber process.

..... [1]

- (f) Ammonia is a weak base.

- (i) Explain the meaning of the term *base*.

.....

..... [1]

- (ii) Suggest the pH of aqueous ammonia.

..... [1]

[Total: 13]

2 Soluble salts can be made by adding a metal carbonate to a dilute acid.

(a) Give the formula of the dilute acid which reacts with a metal carbonate to form a nitrate salt.

..... [1]

(b) A student wanted to make hydrated iron(II) sulfate crystals, $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$, by adding excess iron(II) carbonate to dilute sulfuric acid. The student followed the procedure shown.

step 1 Add dilute sulfuric acid to a beaker.

step 2 Add small amounts of iron(II) carbonate to the dilute sulfuric acid in the beaker until the iron(II) carbonate is in excess.

step 3 Filter the mixture formed in **step 2**.

step 4 Heat the filtrate until it is a saturated solution. Allow to cool.

step 5 Once cold, pour away the remaining solution. Dry the crystals between filter papers.

(i) Why must the iron(II) carbonate be added in excess in **step 2**?

..... [1]

(ii) State **two** observations in **step 2** that would show that iron(II) carbonate was in excess.

1

2 [2]

(iii) Describe what should be done during **step 3** to ensure there is a maximum yield of crystals.

..... [1]

(iv) A saturated solution is formed in **step 4**.

Describe what a saturated solution is.

..... [2]

(v) Name a different compound that could be used instead of iron(II) carbonate to produce hydrated iron(II) sulfate crystals from dilute sulfuric acid.

..... [1]

- (c) On analysing the crystals, the student found that one mole of the hydrated iron(II) sulfate crystals, $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$, had a mass of 278 g.

Determine the value of x using the following steps:

- calculate the mass of one mole of FeSO_4

mass = g

- calculate the mass of H_2O present in one mole of $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$

mass of H_2O = g

- determine the value of x .

x =
[3]

- (d) Insoluble salts can be made by mixing solutions of two soluble salts.

A student followed the procedure shown to make silver bromide, an insoluble salt.

step 1 Add aqueous silver nitrate to a beaker. Then add aqueous potassium bromide and stir.

step 2 Filter the mixture formed in **step 1**.

step 3 Dry the residue.

- (i) State the term used to describe this method of making salts.

..... [1]

- (ii) Give the observation the student would make during **step 1**.

..... [1]

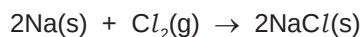
- (iii) Write the ionic equation for the reaction between aqueous silver nitrate and aqueous potassium bromide.

Include state symbols.

..... [3]

- (e) Sodium chloride is an ionic salt. It can be made by reacting sodium with chlorine gas.

The equation for this reaction is shown.



Calculate the volume of chlorine gas, in cm^3 , that reacts to form 2.34 g of NaCl.

The reaction takes place at room temperature and pressure.

volume of chlorine gas = cm^3 [3]

- (f) Sodium chloride does not conduct electricity when solid, but does conduct electricity when molten.

- (i) Explain why, in terms of structure and bonding.

.....
.....
.....
..... [3]

- (ii) Name the product formed at the positive electrode when electricity is passed through molten sodium chloride.

..... [1]

- (iii) State the type of change that occurs at the positive electrode in (ii).

Explain your answer in terms of electron transfer.

type of change
explanation [2]

- (iv) Describe what else can be done to sodium chloride to allow it to conduct electricity.

..... [1]

[Total: 26]

- 4 A student wanted to make some zinc chloride crystals.

The student followed the procedure shown.

step 1 Add excess zinc powder to dilute hydrochloric acid to form aqueous zinc chloride.

step 2 Remove unreacted zinc powder from the aqueous zinc chloride.

step 3 Heat the solution until it is saturated.

step 4 Allow the saturated solution to cool and remove the crystals that form.

- (a) Write the equation for the reaction in **step 1**. Include state symbols.

..... [3]

- (b) Explain why **excess** zinc powder is added in **step 1**.

..... [1]

- (c) Suggest how unreacted zinc powder is removed in **step 2**.

..... [1]

- (d) A saturated solution is formed in **step 3**.

Suggest what is meant by the term *saturated solution*.

.....
..... [2]

- (e) Explain why crystals form as the solution cools in **step 4**.

..... [1]

- (f) Name **two** zinc compounds which react with dilute hydrochloric acid to form zinc chloride.

.....
..... [2]

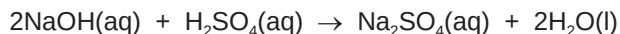
- (g) If excess calcium metal is used instead of excess zinc powder in **step 1**, pure calcium chloride crystals do **not** form.

Explain why.

.....
..... [1]

(h) Some salts can be made by titration.

In a titration experiment, 20.0 cm^3 of aqueous sodium hydroxide reacts exactly with 25.0 cm^3 of 0.100 mol/dm^3 dilute sulfuric acid to make sodium sulfate.



(i) Circle the name of the type of reaction that takes place.

decomposition

neutralisation

precipitation

reduction

[1]

(ii) Calculate the concentration of the aqueous sodium hydroxide in g/dm^3 using the following steps.

- Calculate the number of moles of dilute sulfuric acid used.

..... mol

- Determine the number of moles of sodium hydroxide which react with the dilute sulfuric acid.

..... mol

- Calculate the concentration of the aqueous sodium hydroxide in mol/dm^3 .

..... mol/dm^3

- Calculate the concentration of the aqueous sodium hydroxide in g/dm^3 .

..... g/dm^3
[5]

[Total: 17]

QUESTION 5 STARTS ON THE NEXT PAGE.

1 Give the name of the process that is used:

(a) to produce ammonia from nitrogen

..... [1]

(b) to separate nitrogen from liquid air

..... [1]

(c) to produce bromine from molten lead(II) bromide

..... [1]

(d) to separate an undissolved solid from an aqueous solution

..... [1]

(e) to produce amino acids from proteins

..... [1]

(f) to separate a mixture of amino acids.

..... [1]

[Total: 6]

5 This question is about salts.

- (a) Salts that are soluble in water can be made by the reaction between insoluble carbonates and dilute acids. Zinc sulfate is soluble in water.

You are provided with solid zinc carbonate, ZnCO_3 , and dilute sulfuric acid, H_2SO_4 .

Describe how you would make a pure sample of zinc sulfate crystals.

Your answer should include:

- practical details
- how you would make sure that all the dilute sulfuric acid has reacted
- a chemical equation for the reaction.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

- (b) Some sulfates decompose when heated.

When hydrated iron(II) sulfate is heated strongly, sulfur dioxide gas is given off.

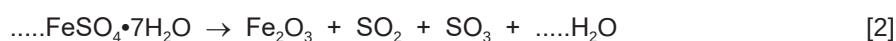
- (i) Describe a test for sulfur dioxide.

test

observations

[2]

- (ii) Complete the equation for the decomposition of hydrated iron(II) sulfate.



(c) Some chlorides are hydrated.

When hydrated barium chloride crystals, $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$, are heated they give off water.



A student carries out an experiment to determine the value of x in $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

step 1 Hydrated barium chloride crystals are weighed.

step 2 The hydrated barium chloride crystals are then heated.

step 3 The remaining solid is weighed.

(i) Describe how the student can be sure that all the water is given off.

.....
.....
..... [2]

(ii) In an experiment, 4.88 g of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ is heated until all the water is given off. The mass of BaCl_2 remaining is 4.16 g.

[M_r : BaCl_2 , 208; H_2O , 18]

Determine the value of x using the following steps.

- Calculate the number of moles of BaCl_2 remaining.

..... mol

- Calculate the mass of H_2O given off.

..... g

- Calculate the number of moles of H_2O given off.

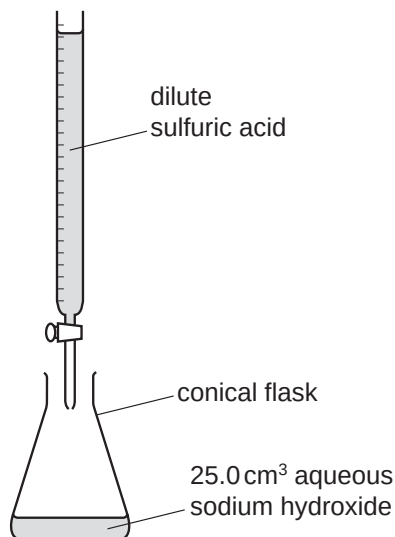
..... mol

- Determine the value of x .

$x =$
[4]

[Total: 15]

- 5 (a) Dilute sulfuric acid and aqueous sodium hydroxide can be used to prepare sodium sulfate crystals using a method that involves titration.



- (i) Suggest why universal indicator is **not** suitable for this titration.

..... [1]

- (ii) Name an indicator that can be used in this titration.

..... [1]

20.0 cm³ of dilute sulfuric acid neutralises 25.0 cm³ of 1.00 mol/dm³ aqueous sodium hydroxide. At the end of the titration the conical flask contains aqueous sodium sulfate with the dissolved indicator as an impurity.

- (b) Describe how to prepare a **pure** sample of sodium sulfate crystals from the original solutions of dilute sulfuric acid and aqueous sodium hydroxide of the same concentrations.

You are not required to give details of how to carry out the titration.

.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [5]

- (c) Sodium hydrogensulfate, NaHSO_4 , dissolves in water to produce an aqueous solution, **X**, containing Na^+ , H^+ and SO_4^{2-} ions.

State the observations when the following tests are done.

- (i) A flame test is carried out on **X**.

..... [1]

- (ii) Copper(II) oxide is warmed with an excess of **X**.

.....
..... [2]

- (iii) Acidified aqueous barium nitrate is added to **X**.

..... [1]

[Total: 11]

6 This question is about zinc and its compounds.

(a) Zinc is extracted from its ore which is mainly zinc sulfide, ZnS.

The steps for this extraction are shown.

step 1 Zinc sulfide is converted into zinc oxide.

step 2 The zinc oxide is then reduced to zinc in a furnace. The zinc formed becomes a gas.

step 3 The zinc gas is cooled to form molten zinc.

(i) Name the ore of zinc, which is mainly zinc sulfide.

..... [1]

(ii) Describe how zinc sulfide is converted into zinc oxide in **step 1**.

.....
..... [1]

(iii) Name the reducing agent used in **step 2**.

..... [1]

(iv) Explain why the zinc forms a gas in **step 2** inside the furnace.

..... [1]

(v) State the name of the physical change occurring when zinc gas is converted into molten zinc.

..... [1]

(b) Zinc sulfate crystals, $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, are hydrated.

Zinc sulfate crystals are made by reacting zinc carbonate with dilute sulfuric acid.

The equation for the overall process is shown.



step 1 Large pieces of solid zinc carbonate are added to dilute sulfuric acid until the zinc carbonate is in excess. This forms aqueous zinc sulfate.

step 2 The excess zinc carbonate is separated from the aqueous zinc sulfate.

step 3 The aqueous zinc sulfate is heated until a saturated solution is formed.

step 4 The saturated solution is allowed to cool and crystallise.

step 5 The crystals are removed and dried.

- (i) In **step 1**, zinc carbonate is in excess when no more zinc carbonate dissolves.

State one **other** observation that indicates the zinc carbonate is in excess in **step 1**.

..... [1]

- (ii) Name a different substance, other than zinc carbonate, that can be added to dilute sulfuric acid to produce aqueous zinc sulfate in **step 1**.

..... [1]

- (iii) **Step 1** is repeated using powdered zinc carbonate instead of large pieces.

All other conditions are kept the same.

The rate of reaction increases.

Give a reason why the rate of reaction increases. Explain your answer in terms of particles.

.....

.....

..... [2]

- (iv) Suggest what is observed when the solution is saturated in **step 3**.

.....

.....

..... [1]

- (v) The formula of zinc sulfate crystals is $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$.

Give the formula of the solid formed if the crystals are heated to dryness in **step 3**.

..... [1]

[Total: 11]

2 Potassium is a Group I element.

(a) Name and describe the bonding in potassium.

name

description

.....

.....

.....

[4]

(b) Potassium combines with sulfur to form an ionic compound, potassium sulfide, K_2S .

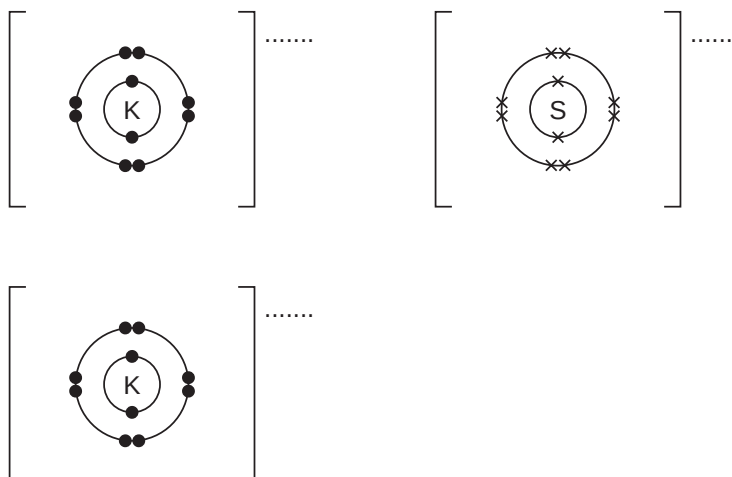
(i) Give **two** physical properties of ionic compounds.

1

2

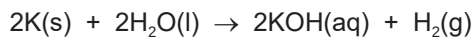
[2]

(ii) Complete the dot-and-cross diagram to show the electron arrangement and charges of the ions in potassium sulfide.



[3]

- (c) When potassium is added to water, it reacts vigorously and a coloured flame is seen. The equation for the reaction is shown.



- (i) State the colour of the flame seen.

..... [1]

- (ii) The solution formed is potassium hydroxide, a strong alkali.

State the formula of the ion responsible for alkalinity in a solution.

..... [1]

- (iii) State the colour of litmus in a strong alkali.

..... [1]

- (iv) Calculate the volume, in cm^3 , of hydrogen gas formed when 2.34 g of potassium is added to excess water at room temperature and pressure.

Use the following steps.

- Calculate the number of moles of potassium added.

= mol

- Determine the number of moles of hydrogen gas formed.

= mol

- Calculate the volume of hydrogen gas formed.

volume = cm^3
[3]

(d) Aqueous potassium hydroxide reacts with a dilute acid to produce aqueous potassium chloride, KCl(aq) , which is a salt.

(i) Name the dilute acid used.

..... [1]

(ii) State the type of reaction taking place.

..... [1]

(iii) Name the experimental technique used when salts are made by reacting a dilute acid with an aqueous alkali.

..... [1]

(e) When aqueous silver nitrate, $\text{AgNO}_3\text{(aq)}$, is added to aqueous potassium chloride, a precipitate is formed.

(i) State the colour of the precipitate formed.

..... [1]

(ii) Name the precipitate formed.

..... [1]

(iii) Write the ionic equation for the reaction. Include state symbols.

..... [3]

[Total: 23]

- 5 Propane, propene, propan-1-ol and propanoic acid are members of different homologous series. Molecules of these substances contain three carbon atoms.

(a) Explain why members of a homologous series have similar chemical properties.

..... [1]

(b) Name the homologous series to which propanoic acid belongs.

..... [1]

(c) State the general formula of the homologous series to which propanoic acid belongs.

..... [1]

(d) Propan-1-ol has an unbranched isomer.

- Name this isomer.

.....

- Draw the displayed formula of this isomer.

[2]

(e) Propane and propene can be manufactured by heating decane, $C_{10}H_{22}$, in the presence of a catalyst. One other product is formed.

(i) Complete the equation for this reaction.

$C_{10}H_{22} \rightarrow$ + + [2]

(ii) Name this manufacturing process.

..... [1]

- (f) Propene forms a polymer named poly(propene).
- (i) Draw the displayed formula of a section of poly(propene) showing **three** repeat units.

[2]

- (ii) State the type of polymerisation that occurs when propene forms poly(propene).

..... [1]

- (g) Propanoic acid reacts with aqueous sodium carbonate to form a salt.

- (i) Suggest the name of the salt formed.

..... [1]

- (ii) Suggest the formula of the anion in this salt.

..... [1]

- (h) Propanoic acid forms an ester when it reacts with ethanol in the presence of a catalyst.

- (i) Suggest a suitable catalyst.

..... [1]

- (ii) Name the ester formed.

..... [1]

- (iii) Draw the displayed formula of this ester.

[2]

[Total: 17]

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lanthanoids	57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	euroium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	ytterbium	173	71	Lu	lutetium	175	
	actinoids	89	Ac	actinium	—	90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium	—	94	Pu	plutonium	—	95	Am	americium	—	96	Cm	curium	—	97	Bk	berkelium	—	98	Cf	californium	—	99	Es	einsteinium	—	100	Fm	fermium	—	101	Md	mendelevium	—	102	No	nobelium	—	103	Lr	lawrencium	—

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

6 This question is about sulfur and compounds of sulfur.

Sulfur is converted into sulfuric acid, H_2SO_4 , by the Contact process.

The process involves four stages.

stage 1 Molten sulfur is converted into sulfur dioxide.

stage 2 Sulfur dioxide reacts with oxygen to form sulfur trioxide.

stage 3 Sulfur trioxide combines with concentrated sulfuric acid to form oleum, $\text{H}_2\text{S}_2\text{O}_7$.

stage 4 Oleum reacts to form concentrated sulfuric acid.

(a) (i) In **stage 1**, iron pyrites, FeS_2 , can be used instead of molten sulfur.
The iron pyrites is heated strongly in air.

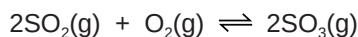
Balance the equation for the reaction occurring when iron pyrites reacts with oxygen in the air.



(ii) Name Fe_2O_3 . Include the oxidation number of iron.

..... [1]

(b) The equation for **stage 2** is shown.



The forward reaction is exothermic.

The reaction is carried out at a temperature of 450°C and a pressure of 2 atm.

Using explanations that do **not** involve cost:

(i) explain why a temperature greater than 450°C is **not** used

.....
..... [1]

(ii) explain why a pressure lower than 2 atm is **not** used.

.....
..... [1]

(c) When sulfuric acid reacts with ammonia the salt produced is ammonium sulfate.

Write the symbol equation for this reaction.

..... [2]

(d) Lead(II) sulfate is an insoluble salt.

Lead(II) sulfate can be made from aqueous ammonium sulfate using a precipitation reaction.

(i) Name a solution that can be added to aqueous ammonium sulfate to produce a precipitate of lead(II) sulfate.

..... [1]

(ii) Write an ionic equation for this precipitation reaction. Include state symbols.

..... [3]

(iii) The precipitate of lead(II) sulfate forms in an aqueous solution.

Describe how pure lead(II) sulfate can be obtained from the mixture.

.....
.....
..... [3]

[Total: 13]

4 Solid sodium hydroxide is a base which dissolves to form an aqueous solution, NaOH(aq).

(a) State what is meant by the term base.

..... [1]

(b) State the term given to a base which dissolves to form an aqueous solution.

..... [1]

(c) State the colour of thymolphthalein in NaOH(aq).

..... [1]

(d) Complete the word equation for the reaction of NaOH(aq) with ammonium chloride.

sodium + ammonium → + +
hydroxide chloride
.....

[3]

(e) Some metal oxides react with NaOH(aq).

(i) State the term given to metal oxides which react with bases such as NaOH(aq).

..... [1]

(ii) Name a metal oxide which reacts with NaOH(aq).

..... [1]

- (f) Ethanoic acid, CH_3COOH , is a weak acid.
- (i) Complete the dot-and-cross diagram in Fig. 4.1 of a molecule of ethanoic acid.

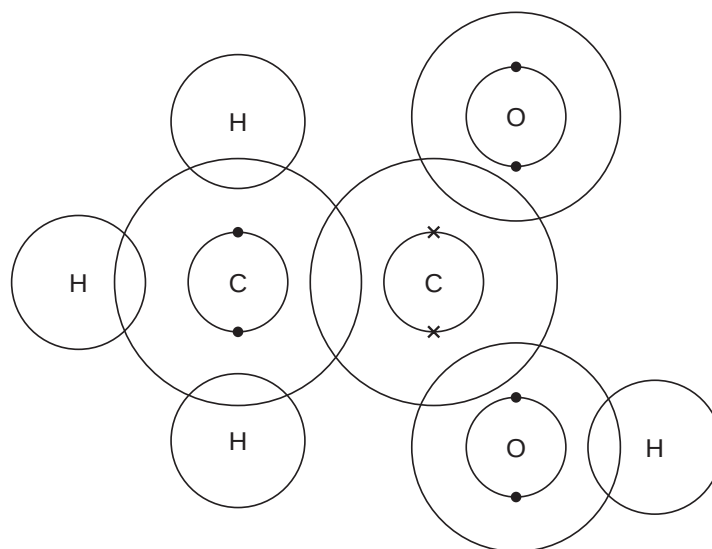


Fig. 4.1

[3]

- (ii) Suggest the pH of dilute ethanoic acid.

..... [1]

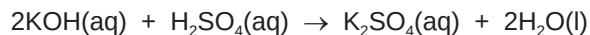
- (iii) Complete the symbol equation to show the dissociation of ethanoic acid.

CH_3COOH [3]

- (iv) Write the ionic equation for the reaction when an acid neutralises a soluble base.

..... [1]

- (g) In a titration, 25.0 cm³ of 0.0800 mol/dm³ aqueous potassium hydroxide, KOH(aq), is neutralised by 20.0 cm³ of dilute sulfuric acid, H₂SO₄(aq).



Calculate the concentration of H₂SO₄, in g/dm³ using the following steps.

- Calculate the number of moles of KOH used.

..... mol

- Determine the number of moles of H₂SO₄ which react with the KOH.

..... mol

- Calculate the concentration of H₂SO₄ in mol/dm³.

..... mol/dm³

- Calculate the concentration of H₂SO₄ in g/dm³.

..... g/dm³
[5]

[Total: 21]

- 1 A list of substances is shown.

barium nitrate
carbon monoxide
hydrated cobalt(II) chloride
copper(II) oxide
anhydrous copper(II) sulfate
ethane
potassium iodide
propene
sodium bromide
sulfur dioxide
zinc oxide

Answer the following questions using only the substances from the list.
Each substance may be used once, more than once or not at all.

Give the name of the substance that:

- (a) gives a lilac colour in a flame test

..... [1]

- (b) forms a cream precipitate when its aqueous solution reacts with acidified aqueous silver nitrate

..... [1]

- (c) is an acidic oxide

..... [1]

- (d) is an unsaturated hydrocarbon

..... [1]

- (e) is a product of incomplete combustion of fossil fuels

..... [1]

- (f) is used to test for the presence of water.

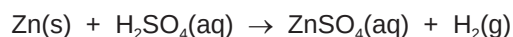
..... [1]

[Total: 6]

4 This question is about sulfuric acid and salts that are made from sulfuric acid.

(a) Zinc reacts with dilute sulfuric acid. Aqueous zinc sulfate is one of the products.

Powdered zinc is added to dilute sulfuric acid. The mixture is stirred. More zinc is added, with stirring, until the zinc is in excess.



The mixture is then filtered.

(i) Name the limiting reactant.

..... [1]

(ii) State two **observations** that indicate the zinc is in excess.

1

2 [2]

(iii) Name the filtrate.

..... [1]

(iv) Name **two** compounds which both react with dilute sulfuric acid to produce aqueous zinc sulfate.

1

2 [2]

(b) Zinc sulfate crystals are produced by heating aqueous zinc sulfate until a saturated solution is formed. When the saturated solution cools down, crystals of zinc sulfate start to form.

(i) State what is meant by the term saturated solution.

.....

..... [2]

(ii) Explain why crystals form when the saturated solution cools down.

..... [1]

(c) Nickel(II) sulfate crystals contain water of crystallisation.

When nickel(II) sulfate crystals, $\text{NiSO}_4 \cdot x\text{H}_2\text{O}$, are heated, they give off water.



A student carries out an experiment to determine the value of x in $\text{NiSO}_4 \cdot x\text{H}_2\text{O}$.

step 1 Nickel(II) sulfate crystals are weighed.

step 2 Nickel(II) sulfate crystals are heated.

step 3 The remaining solid is allowed to cool and is then weighed.

step 4 The remaining solid is heated again, allowed to cool and is then weighed.

step 5 Step 4 is repeated until there is no change in mass.

(i) State the term used to describe crystals that contain water of crystallisation.

..... [1]

(ii) State why **step 4** is repeated until there is no change in mass.

..... [1]

(iii) In an experiment, 0.454 g of nickel(II) sulfate crystals, $\text{NiSO}_4 \cdot x\text{H}_2\text{O}$, is used. The mass of anhydrous nickel(II) sulfate, NiSO_4 , remaining is 0.310 g.

[M_r : NiSO_4 , 155; H_2O , 18]

Determine the value of x in $\text{NiSO}_4 \cdot x\text{H}_2\text{O}$.

Use the following steps.

- Calculate the number of moles of NiSO_4 remaining.

moles of NiSO_4 =

- Calculate the mass of H_2O given off.

mass of H_2O = g

- Calculate the number of moles of H_2O given off.

moles of H_2O =

- Calculate the value of x .

$x = \dots\dots\dots$
[4]

[Total: 15]

- 3 This question is about acids, bases and alkalis.

Table 3.1 shows the pH values of some substances.

Table 3.1

substance	pH
NaOH(aq)	14
Ca(OH) ₂ (aq)	10
H ₂ O(l)	7
CH ₃ COOH(aq)	4
HNO ₃ (aq)	1

- (a) Define the term base.

..... [1]

- (b) State what is meant by the term alkali.

..... [1]

- (c) Thymolphthalein is an indicator.

State the colour of thymolphthalein in:

- NaOH(aq)
- CH₃COOH(aq).

[2]

- (d) (i) Use the information in Table 3.1 to identify the substance with the highest concentration of H⁺(aq) ions.

Explain your answer.

substance

explanation

[2]

- (ii) Name an indicator which can be used to identify the substance with the highest concentration of H⁺(aq) ions.

..... [1]

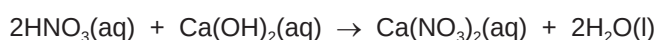
- (e) Complete the equation to show the dissociation of ethanoic acid, CH_3COOH , in aqueous solution.

$\text{CH}_3\text{COOH}(\text{aq})$ [3]

- (f) Write the **ionic** equation which represents a neutralisation reaction between any acid and any alkali.

..... [1]

- (g) Dilute nitric acid, $\text{HNO}_3(\text{aq})$, reacts with aqueous calcium hydroxide, $\text{Ca}(\text{OH})_2(\text{aq})$, as shown.



20.0 cm^3 of 0.0150 mol/dm^3 $\text{Ca}(\text{OH})_2(\text{aq})$ reacts with 25.0 cm^3 of $\text{HNO}_3(\text{aq})$.

Calculate the concentration of $\text{HNO}_3(\text{aq})$ in g/dm^3 .

Use the following steps.

- Calculate the number of moles of $\text{Ca}(\text{OH})_2(\text{aq})$ used.

..... mol

- Determine the number of moles of $\text{HNO}_3(\text{aq})$ which react with the $\text{Ca}(\text{OH})_2(\text{aq})$.

..... mol

- Calculate the concentration of $\text{HNO}_3(\text{aq})$ in mol/dm^3 .

..... mol/dm^3

- Calculate the concentration of $\text{HNO}_3(\text{aq})$ in g/dm^3 .

..... g/dm^3
[5]

[Total: 16]

5 (a) Barium sulfate, BaSO_4 , is an insoluble salt and is made by precipitation.

(i) Name **two** aqueous solutions that produce a precipitate of barium sulfate when they are mixed.

1

2 [2]

(ii) Describe how to produce a pure sample of barium sulfate from the mixture of aqueous solutions in (a)(i).

.....

..... [2]

(iii) Write an ionic equation for the precipitation reaction which produces barium sulfate. Include state symbols.

..... [3]

(b) Soluble salts are made from dilute acids.

Name the dilute acid and one other substance that react together to make copper(II) sulfate.

dilute acid

other substance [2]

(c) Nitrates decompose when they are heated.

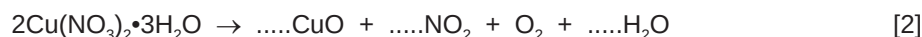
When hydrated copper(II) nitrate is heated, oxygen gas is produced.

(i) Describe a test for oxygen.

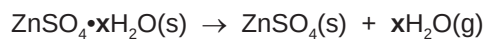
test

observations [1]

(ii) Complete the equation for the decomposition of hydrated copper(II) nitrate.



(d) Hydrated zinc sulfate gives off water when it is heated.



A student does an experiment to determine the value of x in $\text{ZnSO}_4 \cdot x\text{H}_2\text{O}$.

step 1 The student weighs a sample of hydrated zinc sulfate.

step 2 The student heats the sample of hydrated zinc sulfate.

step 3 The student weighs the solid after heating.

step 4 The student repeats **step 2** and **step 3** until the mass of solid after heating is constant.

(i) State why the student does **step 4**.

..... [1]

(ii) In an experiment, 0.574 g of $\text{ZnSO}_4 \cdot x\text{H}_2\text{O}$ is heated until the mass is constant.

The mass of ZnSO_4 that remains is 0.322 g.

[M_r : ZnSO_4 , 161; H_2O , 18]

Determine the value of x using the following steps.

- Calculate the number of moles of ZnSO_4 remaining.

..... mol

- Calculate the mass of H_2O given off.

..... g

- Calculate the number of moles of H_2O given off.

..... mol

- Determine the value of x .

$x =$ [4]

[Total: 17]

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1 A list of chemical and physical processes, **A** to **H**, is shown.

- A** combustion
- B** diffusion
- C** melting
- D** neutralisation
- E** photosynthesis
- F** reversible reaction
- G** roasting
- H** thermal decomposition

Answer the following questions about processes **A** to **H**.
Each letter may be used once, more than once or not at all.

State which of the processes **A** to **H**:

(a) happens when an acid reacts with an alkali

..... [1]

(b) reaches a position of equilibrium

..... [1]

(c) involves particles changing from fixed positions to being mobile, but still touching

..... [1]

(d) are physical changes

..... and [1]

(e) is caused by gas particles colliding with each other.

..... [1]

[Total: 5]



6 Natural polyamides are polymers made from amino acid monomers.

(a) State the type of polymerisation reaction that occurs when natural polyamides form.

..... [1]

(b) State the term given to natural polyamides.

..... [1]

(c) An amino acid is represented as shown in Fig. 6.1.



Fig. 6.1

Complete Fig. 6.2 to show the general structure of an amino acid.

Show all of the atoms and all of the bonds in the functional groups.



Fig. 6.2

[3]





(d) Three different amino acids are represented as shown in Fig. 6.3.

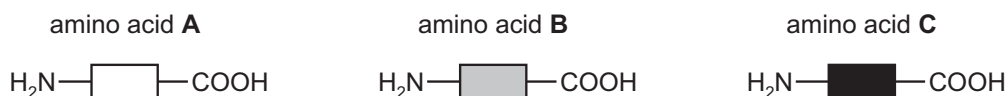


Fig. 6.3

Complete the diagram in Fig. 6.4 to show the part of the structure of the natural polyamide that forms when the three amino acids, **A**, **B** and **C**, combine.
Show all of the atoms and all of the bonds in the linkages.



Fig. 6.4

[3]

(e) A mixture of the three amino acids, **A**, **B** and **C**, can be separated and the amino acids identified using paper chromatography.

Complete the equation for R_f .

$$R_f =$$

[2]





- (f) A sample of the mixture of the three amino acids, **A**, **B** and **C**, is placed onto the baseline and a chromatogram is allowed to develop as shown in Fig. 6.5.

The finished chromatogram is shown in Fig. 6.6.

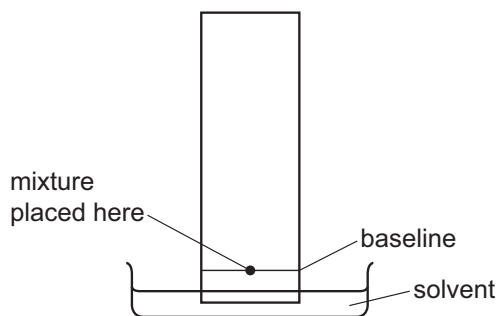


Fig. 6.5



Fig. 6.6

The amino acids, **A**, **B** and **C**, are colourless. Water is used as the solvent.

- (i) Explain why the baseline is drawn in pencil.

..... [1]

- (ii) State the type of substance used to make the colourless amino acids visible on the chromatogram in Fig. 6.6.

..... [1]

- (iii) Explain why in Fig. 6.6 only **two** spots are seen from the mixture of three amino acids.

.....
..... [1]

- (iv) Suggest how the experiment can be changed to separate all three amino acids.

..... [1]

[Total: 14]



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The Periodic Table of Elements

Group																																				
I	II																VIII																			
lanthanoids	3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass														1 H hydrogen 1																	2 He helium 4		
	11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84										
	37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xeron 131	55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
	87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —																		
	lanthanoids																																			
	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175																					
	actinoids																																			
	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —																					

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

4 This question is about sulfuric acid, H_2SO_4 .

- (a) Dilute sulfuric acid and aqueous sodium hydroxide can be used to prepare sodium sulfate crystals using a method that involves titration.

The apparatus for titration is shown in Fig. 4.1.

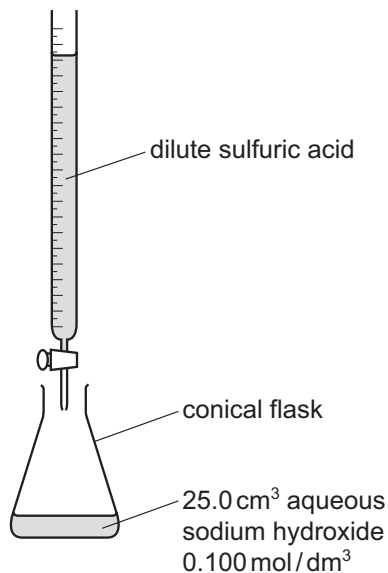


Fig. 4.1

Thymolphthalein is used as an indicator for this titration.

- (i) State the colour change of thymolphthalein at the end-point of this titration.

from to [2]

- (ii) Suggest why universal indicator is **not** used for this titration.

.....
..... [1]



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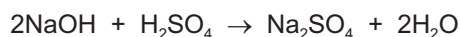
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- (b) 25.0 cm³ of aqueous sodium hydroxide, NaOH, of concentration 0.100 mol/dm³ is neutralised by 20.0 cm³ of dilute sulfuric acid, H₂SO₄.

The equation for the reaction is shown.



Calculate the concentration of H₂SO₄ using the following steps.

- Calculate the number of moles of NaOH used.

..... mol

- Determine the number of moles of H₂SO₄ that react with the NaOH.

..... mol

- Calculate the concentration of H₂SO₄.

..... mol/dm³
[3]

- (c) A student is provided with an aqueous solution of sodium sulfate.

Describe how to prepare a **pure** sample of sodium sulfate crystals from this solution.

.....

 [3]

- (d) Potassium hydrogen sulfate, KHSO₄, can be prepared by a reaction between aqueous potassium hydroxide and dilute sulfuric acid. Water is the only other product.

Write a symbol equation for this reaction.

..... [1]





(e) Potassium hydrogen sulfate, KHSO_4 , dissolves in water to form solution **X**.

Solution **X** contains K^+ , H^+ and SO_4^{2-} ions.

(i) Name the **type** of solution that contains H^+ ions.

..... [1]

(ii) State the observations when the following tests are done.

- A flame test is carried out on **X**.

.....

- Solid copper(II) carbonate is added to **X**.

.....

.....

.....

- Aqueous barium nitrate acidified with dilute nitric acid is added to **X**.

.....

[5]

(f) 0.325 g of Zn is added to dilute sulfuric acid which contains 0.0100 moles of H_2SO_4 .

The equation for this reaction is shown.



(i) Determine whether Zn or H_2SO_4 is the limiting reactant.
Explain your answer.

.....

.....

.....

..... [2]



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- (ii) In another experiment, 48.0 cm^3 of hydrogen gas, H_2 , is produced. The experiment is carried out at room temperature and pressure, r.t.p.

Calculate the number of molecules in 48.0 cm^3 of H_2 gas measured at r.t.p.

The value of the Avogadro constant is 6.02×10^{23} .

..... molecules [2]

[Total: 20]



5 This question is about rate of reaction and equilibrium.

A student investigates the rate of decomposition of aqueous hydrogen peroxide, H_2O_2 , using manganese(IV) oxide as a catalyst.

The equation for the reaction is shown.



The student uses the apparatus shown in Fig. 5.1.

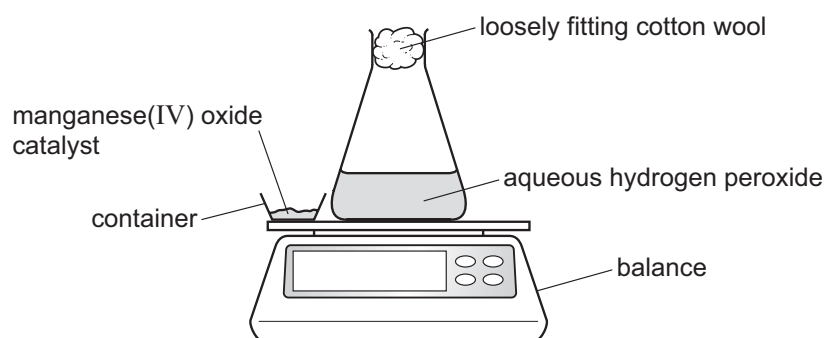


Fig. 5.1

The student:

- adds the catalyst to the aqueous hydrogen peroxide
- replaces the container on the balance
- starts a stop-watch
- records the mass at regular time intervals.

(a) Table 5.1 shows the mass recorded at regular time intervals.

Table 5.1

time / s	mass / g
0	50.64
30	49.80
60	49.38
90	49.17
120	49.07
150	49.02
180	48.99
210	48.97
240	48.97
270	48.97

(i) Suggest why the mass decreases as time increases.

[1]



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- (ii) After a certain time the reaction stops.

Explain why the reaction stops.

..... [1]

- (iii) Suggest why it is **not** possible to use the results in Table 5.1 to determine the **exact** time when the reaction stops.

.....

..... [1]

- (b) Fig. 5.2 shows a graph of the mass against time.

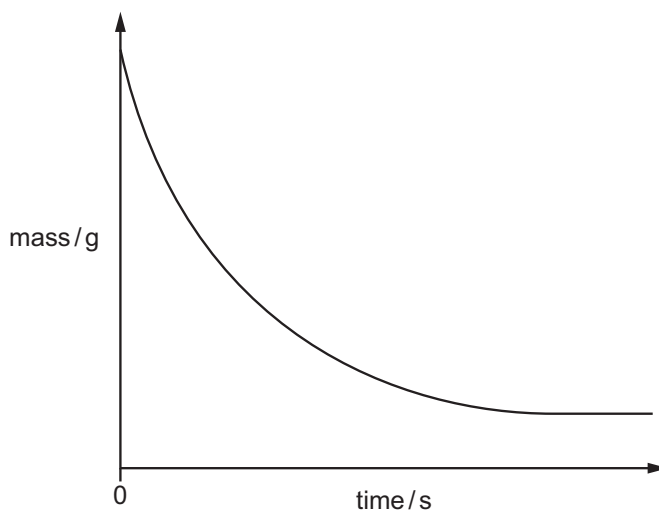


Fig. 5.2

The experiment is repeated at a higher temperature.
All other conditions remain the same.

- (i) Explain, in terms of collision theory, why the rate of reaction is higher at a higher temperature.

.....

.....

.....

.....

..... [3]

- (ii) On Fig. 5.2, sketch the line expected when the experiment is repeated at a higher temperature. [2]





(c) Manganese(IV) oxide is the catalyst in this reaction.

(i) Explain the meaning of (IV) in manganese(IV) oxide.

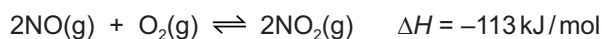
.....
 [2]

(ii) State how the mass of the **catalyst** has changed, if at all, at the end of the experiment.

..... [1]

(d) Nitrogen monoxide gas, NO, and oxygen gas, O₂, react to produce nitrogen dioxide gas, NO₂, at room temperature.

The reaction can reach equilibrium. The equation is shown.



NO(g) and O₂(g) are passed into a beaker as shown in Fig. 5.3.

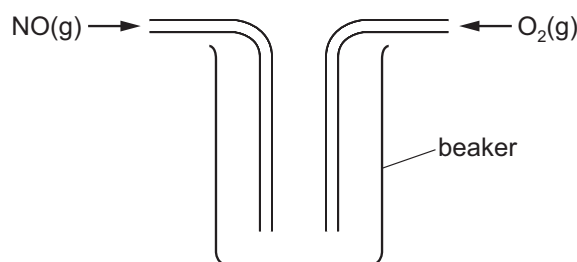


Fig. 5.3

(i) Explain why the method shown in Fig. 5.3 will **not** allow the reaction to reach equilibrium.

..... [1]

(ii) The apparatus is changed and equilibrium is reached.

The temperature of the equilibrium system is then increased and the position of equilibrium shifts to the left.

Explain why the position of equilibrium shifts to the left.

..... [1]



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(iii) The pressure of the equilibrium system is then increased.

State the direction, if any, in which the position of equilibrium shifts.
Explain your answer.

direction

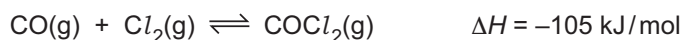
explanation

[2]

[Total: 15]



- 4 Carbonyl chloride, COCl_2 , is manufactured by reacting carbon monoxide with chlorine.



The process takes place in a closed system, and an equilibrium is reached.
The conditions for this process are 200 °C and 200 kPa.

- (a) Explain what is meant by the term closed system.

..... [1]

- (b) State what the symbol ΔH represents.

..... [1]

- (c) State how the value of ΔH shows that the forward reaction is exothermic.

..... [1]

- (d) Deduce the value of ΔH for the reverse reaction. Include a sign in your answer.

..... [1]

- (e) Complete Table 4.1 to show the effect, if any, on the concentration of $\text{COCl}_2\text{(g)}$ at equilibrium when the following changes to the conditions are applied.

Use only the words **increases**, **decreases** or **no change**.

Table 4.1

change to conditions	effect on the concentration of $\text{COCl}_2\text{(g)}$ at equilibrium
the temperature is increased	
some CO is added	
the pressure is increased	
a catalyst is added	

[4]





(f) The equation for the reaction can be represented as shown in Fig. 4.1.

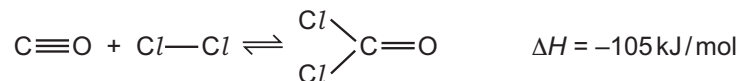


Fig. 4.1

Table 4.2 shows some bond energies.

Table 4.2

bond	$\text{C}\equiv\text{O}$	$\text{Cl}-\text{Cl}$	$\text{C}-\text{Cl}$
bond energy in kJ/mol	1075	240	340

Use the bond energies in Table 4.2 and the value of ΔH for the reaction to calculate the bond energy, in kJ/mol, of the $\text{C}=\text{O}$ bond.

Use the following steps.

- Calculate the energy needed to break the bonds in the reactants.

..... kJ

- Calculate the energy released when the bonds in carbonyl chloride form.

..... kJ

- Calculate the bond energy of the $\text{C}=\text{O}$ bond.

..... kJ/mol
[3]



(g) Complete the dot-and-cross diagram in Fig. 4.2 of a molecule of carbonyl chloride.

Show outer shell electrons only.

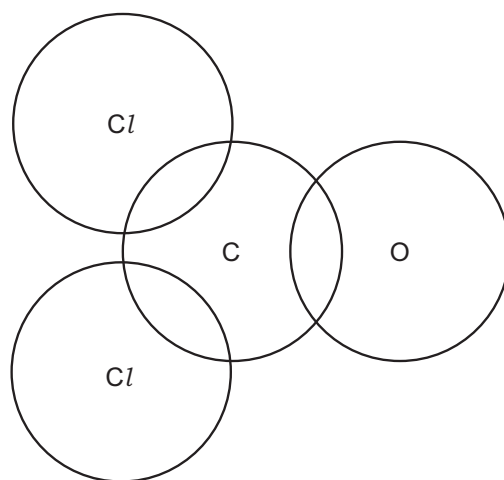


Fig. 4.2

[3]

[Total: 14]



3 This question is about zinc and compounds of zinc.

(a) Zinc is held together by electrostatic forces of attraction between particles.

(i) Name the type of bonding in zinc.

..... [1]

(ii) Name the **two** types of particles held together by the bonding in (a)(i).

1

2 [2]

(iii) Name the type of particle whose movement allows zinc to conduct electricity.

..... [1]

(b) Zinc is present in alloys such as brass.

(i) State the meaning of the term alloy.

..... [1]

(ii) Name the substance that is present in brass, other than zinc.

..... [1]

(c) Zinc sulfate crystals are made by the reaction between zinc carbonate and dilute sulfuric acid, using the following steps.

step 1 An excess of powdered zinc carbonate is added to dilute sulfuric acid.

step 2 Excess zinc carbonate is separated from aqueous zinc sulfate by filtration.

step 3 Aqueous zinc sulfate is heated until a saturated solution is formed.

step 4 The saturated solution is allowed to cool and crystallise.

step 5 The crystals are removed and dried.

(i) Give **two** observations which show that the zinc carbonate is in excess in **step 1**.

1

2 [2]

(ii) Name the filtrate in **step 2**.

..... [1]



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- (iii) Name a compound, other than zinc carbonate, that can be added to dilute sulfuric acid to produce aqueous zinc sulfate in **step 1**.

..... [1]

- (iv) State what is meant by the term saturated solution.

.....

..... [2]

- (v) **Step 1** is repeated using large pieces of zinc carbonate instead of powdered zinc carbonate.

All other conditions are the same.

The rate of reaction decreases.

Explain why the rate of reaction decreases. Give your answer in terms of particles.

.....

.....

..... [2]

- (vi) Hydrated crystals form in **step 4**.

State what is meant by the term hydrated.

..... [1]

[Total: 15]



4 Nitric acid, HNO_3 , and sulfuric acid, H_2SO_4 , are strong acids. Ethanoic acid, CH_3COOH , is a weak acid.

(a) One mole of each acid is added separately to 1000cm^3 of distilled water to form a dilute solution. All three acids dissociate in distilled water.

(i) State the formula of the common cation produced when each of the three acids dissociates in water.

..... [1]

(ii) Explain, in terms of dissociation, why:

- nitric acid is described as a strong acid

.....

- ethanoic acid is described as a weak acid.

.....

[2]

(iii) State which of the three dilute acids will have the highest pH value.

..... [1]

(iv) State the colour of thymolphthalein in all three dilute acids.

..... [1]

(v) Give the formula of the anion formed when:

- sulfuric acid dissociates in water

.....

- ethanoic acid dissociates in water.

.....

[2]

(vi) Name the **two** gaseous products formed during the electrolysis of dilute sulfuric acid using inert electrodes.

..... and [2]

(vii) Name the salt formed when calcium reacts with ethanoic acid.

..... [1]



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(b) Aluminium nitrate, $Al(NO_3)_3$, is made when dilute nitric acid reacts with aluminium hydroxide, $Al(OH)_3$.

(i) $Al(OH)_3$ acts as a base in this reaction.

Define the term base.

..... [1]

(ii) $Al(OH)_3$ is **not** an alkali.

State what this tells you about the solubility of $Al(OH)_3$.

..... [1]

(iii) Name **one other** compound which forms aluminium nitrate when added to dilute nitric acid.

..... [1]

(iv) The oxidation number of the O atoms in $Al(NO_3)_3$ is -2 .

The formula of the aluminium ion in $Al(NO_3)_3$ is Al^{3+} .

Determine the oxidation number of the N atoms in $Al(NO_3)_3$.
Show your working.

oxidation number of N = [2]

[Total: 15]



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5 This question is about metallic compounds.

(a) Insoluble metal salts, such as lead(II) sulfate, are made by precipitation.

When aqueous lead(II) ions react with aqueous sulfate ions, a precipitate of lead(II) sulfate forms.

A sample of pure lead(II) sulfate can be obtained from the precipitate formed.

(i) Write an ionic equation for this precipitation reaction. Include state symbols.

..... [3]

(ii) State the name of an aqueous solution containing lead(II) ions.

..... [1]

(iii) State the name of an aqueous solution containing sulfate ions.

..... [1]

(iv) The lead(II) sulfate precipitate is separated from the rest of the reaction mixture by filtration. The impure lead(II) sulfate remains on the filter paper.

State the general term given to an insoluble substance that remains on the filter paper during filtration.

..... [1]

(v) Describe how the lead(II) sulfate is purified after filtration.

.....
..... [2]





(b) Copper is a transition element.

A list of properties of copper is shown.

- It has good electrical conductivity.
- It has variable oxidation states.
- It forms soluble salts.
- It forms coloured compounds.
- It forms basic oxides.
- It acts as a catalyst.

(i) Give **two** properties from the list that show how copper is similar to metals in Group I of the Periodic Table.

1

2 [2]

(ii) Give **two** properties from the list that show how copper is different from metals in Group I of the Periodic Table.

1

2 [2]

[Total: 12]

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